

▲ Builder Basics · First Pyramid — English Worksheet

Topic: First Pyramid · **Course:** Builder Basics · **Time:** about 45 minutes

The agent **walks the edges** of a square, dropping blocks as it moves. Stack the squares, shrink each one, and a pyramid grows.

1 · Predict 🧠

Read each set of steps. Write what you think will happen.

```
set f to 6
place on move ON
repeat 4 times:
  move forward by f
  turn left
```

What shape does the agent draw? How many blocks long is each side?

Why is it **repeat 4 times** and **turn left** ? What would **repeat 3 times** draw instead?


```
set f to 6
repeat 3 times:
  [draw one f × f square]
  move up by 1
  change f by -2
```

List the side length of each layer from bottom to top. Looking from the side, what shape do the stacked squares make?


```
place on move ON  
move forward by 4  
place on move OFF  
move forward by 4
```

Where do the blocks end up? Why does turning **place on move** OFF matter when the agent moves up to start the next layer?

2 · Spot the Bug 🐛

Each code block below is broken. Rewrite it so it works, then explain the fix.

Bug A — The agent is supposed to draw a closed **square** outline.

```
set f to 6
place on move ON
repeat 4 times:
  move forward by f
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Each new layer must be **2 blocks smaller** than the one below, so the tower narrows into a pyramid.

```
set f to 8
repeat 4 times:
  [draw one f × f square]
  move up by 1
  change f by 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Each square must sit one block **higher** than the last.

```
set f to 8
repeat 4 times:
  [draw one f × f square]
  change f by -2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Run `pyra`, change the **starting value of `f`** to build a shorter pyramid, then reset a stuck agent with `1` (left), `r` (right), `r1` (back).

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your code. Scroll through it. Say what each part does.

2 • Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: ____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit

Send this worksheet + your video to teacher on KakaoTalk.